



TUM School of Management

Campus Challenge: Successful Investing with AI and Large Language Models

-Report-

Report submitted by:

Yamama Al Mghawish (03796819)

E-mail: yamama.al@mytum.de

Yesenia Marin Paniagua (03796704)

E-mail: yesenia.marin@tum.de

Kexin Zhao (03797007)

E-mail: kexin.zhao@tum.de

Maximilian-Luca Seidlitz (03824402)

E-mail: maximilian.seidlitz@tum.de

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Prof. Dr. Sebastian Müller

Professorship of Finance

TUM School of Management

TUM Campus Heilbronn

Technical University of Munich

Bildungscampus 9, 74076 Heilbronn

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List of abbreviations

AI	Artificial Intelligence
CMA	Conservative Minus Aggressive (investing factor)
ESG	Environmental, Social and Governance
HAC	Heteroskedasticity- and autocorrelation-consistent
HML	High Minus Low (value factor)
LLM	Large Language Model
NLP	Natural Language Processing
OLS	Ordinary Least Squares
RMW	Robust Minus Weak (profitability factor)
SMB	Small Minus Big (size factor)

1 Introduction

Recent advances in large language models (LLMs) have generated growing interest in their potential applications in financial markets, particularly for processing unstructured textual information such as financial news headlines. Asset prices react to large volumes of qualitative information every day, yet the extent to which this information is efficiently incorporated into prices remains an open question. Building on this, recent work by Lopez-Lira and Tang (2023) provides early evidence that sentiment generated by LLMs can be used to forecast stock returns and construct profitable trading strategies. A key implication of their findings is that the performance of such strategies depends not only on the underlying language model, but also on how sentiment is extracted, especially on the design of the prompt used to guide the model's output.

Motivated by this insight, the objective of this study is to revisit the sentiment-based trading strategy proposed by Lopez-Lira and Tang (2023) by designing and evaluating an alternative sentiment analysis prompt. Rather than developing a new model or dataset, the focus lies on prompt optimization and systematic performance evaluation, in line with the project assignment. The central research question is whether modifying the sentiment prompt affects portfolio performance and risk-adjusted returns, and whether LLM-generated sentiment remains economically meaningful under this revised implementation.

The study is situated within the broader literature on sentiment, machine learning, and asset pricing, which examines how textual information influences investor behavior and market outcomes. While earlier approaches relied on dictionary-based sentiment measures and supervised machine learning techniques, LLMs represent a more recent development due to their ability to capture contextual and semantic aspects of language without task-specific training. Accordingly, the analysis evaluates whether prompt-based sentiment signals derived from financial news headlines can be used to construct long-only and long-short trading strategies, and how these strategies perform once standard risk factors are taken into account.

Empirically, the study uses a dataset of financial news headlines and corresponding stock returns provided as part of the project. Sentiment is generated using the GPT-4.1-mini language model together with a prompt that is optimized within this study. Based on this sentiment, trading portfolios are constructed using a daily rebalancing and equal-weighting scheme.

Portfolio performance is assessed using cumulative and annualized returns, volatility, and Sharpe ratios. To evaluate whether observed returns represent abnormal performance, the analysis applies the Capital Asset Pricing Model as well as Fama–French three-factor and five-factor models, with heteroscedasticity- and autocorrelation-robust standard errors.

The report proceeds by first reviewing the relevant literature on sentiment measures, machine learning, and large language models in asset pricing, with particular attention to the framework of Lopez-Lira and Tang (2023). It then describes the data, pre-processing steps, and the design and selection of the sentiment prompt. Next, the empirical methodology is outlined, followed by the presentation of the empirical results. The findings are subsequently discussed and compared with prior research, and the report concludes with a summary of the main insights and their implications.

2 Literature Review

2.1 Traditional Sentiment Measures in Finance

Early research in finance demonstrates that investor sentiment plays a systematic role in asset price formation, challenging the strong form of market efficiency. While classical asset pricing models assume that prices fully and instantaneously reflect all available information (Fama, 1970), a growing behavioral finance literature documents persistent deviations from this benchmark attributable to bounded rationality, biased beliefs, and limits to arbitrage (Barberis, Shleifer, & Vishny, 1998; Shleifer & Vishny, 1997). Within this framework, sentiment, defined as investors' beliefs or emotions not fully justified by fundamentals, has been shown to affect both stock returns and trading behavior. A seminal contribution by Tetlock (2007) demonstrates that the tone of media coverage contains economically meaningful information, with high levels of pessimism predicting downward pressure on market returns and increased volatility. Following studies extend this insight by showing that negative or sensational news content leads to temporary mispricing that reverses over time, consistent with investor underreaction and gradual information diffusion (Tetlock, Saar-Tsechansky, & Macskassy, 2006). More broadly, these findings suggest that textual information embedded in news media captures dimensions of investor sentiment that are not immediately incorporated into prices, thereby generating short-term return predictability. Collectively, this literature establishes sentiment as a relevant state variable in financial markets and provides the conceptual foundation for later empirical

approaches that seek to quantify sentiment using increasingly sophisticated textual analysis techniques.

2.2 Machine Learning and NLP in Asset Pricing

Building on early evidence that textual sentiment affects asset prices, later research increasingly employs machine learning and natural language processing (NLP) techniques to extract information from large-scale text data in a systematic and replicable manner. Traditional dictionary-based approaches, while influential, face limitations in capturing context, negation, and domain-specific language usage. In response, Loughran and McDonald (2010) demonstrate that finance-specific word classifications substantially outperform generic sentiment dictionaries, emphasizing the importance of tailored textual analysis in financial applications. More recent studies extend this line of work by leveraging supervised and unsupervised machine learning models to process high-dimensional textual data, allowing for richer representations of economic information embedded in corporate disclosures and news articles. For instance, Boudoukh et al. (2019) and Kelly, Pruitt, and Su (2018) show that machine learning methods can identify latent patterns in textual and firm-level characteristics that improve return predictability beyond traditional linear factor models. Similarly, Gu, Kelly, and Xiu (2019) document that nonlinear machine learning algorithms outperform classical asset pricing models in forecasting the cross-section of stock returns, particularly when applied to large and complex datasets. This literature establishes machine learning and NLP as powerful tools for asset pricing research, enabling researchers to move beyond manually constructed sentiment measures toward data-driven representations of information that capture subtle and nonlinear relationships between text and expected returns.

2.3 Large Language Models in Financial Prediction

The impact of Large Language Models (LLMs) on financial forecasting is substantial and increasingly recognized in recent literature. As the capabilities of models evolve, a growing number of use cases are emerging within the financial sector. Consequently, the application of LLMs for financial prediction is progressively replacing traditional deep learning approaches, such as BERT (Lopez-Lira & Tang, 2023).

In the domain of sentiment analysis, the advanced contextual understanding of modern LLMs offers a significant advantage in evaluating news and headlines (Nie et al., 2024). Unlike older models, LLMs such as GPT-4 can interpret irony, vague or ambiguous statements from

institutions, and complex causal chains (Nie et al., 2024). Lopez-Lira and Tang (2023) define this competency as an "emergent capacity" for financial reasoning. They document that while smaller models fail to generate significant predictive signals, larger models like GPT-4 can effectively synthesize unstructured data to predict stock price movements, outperforming domain-specific models trained on smaller datasets.

Expanding on this technological foundation, Nie et al. (2024) categorize the transformative impact of LLMs into several distinct domains. Beyond standard text processing, they emphasize the automation of linguistic tasks, such as distilling Environmental, Social and Governance (ESG) indicators from corporate disclosures. In the realm of sentiment analysis, LLMs offer a qualitative leap over lexicon-based approaches by decoding the semantic nuances of financial news, a critical factor for accurate prediction. Furthermore, the authors highlight the models' capacity for financial reasoning, which enables the resolution of intricate analytical problems like balance sheet interpretation. Perhaps most innovatively, the survey points to agent-based modeling, where LLMs simulate autonomous market participants to study complex crowd dynamics and price formation mechanisms. From a theoretical perspective, the suitability of LLMs for asset pricing is driven by their ability to model non-linear market dynamics. Traditional asset pricing models, such as the Capital Asset Pricing Model (CAPM) and Fama-French factors, rely on linear relationships that often fail to fully capture the intricacies of modern financial markets. Specifically, the complex interactions and collective reactions of market participants are crucial for accurate forecasting - an area where LLMs demonstrate particular strength. Nie et al. (2024) argue that deep learning models can identify these non-linear patterns, thereby establishing an information edge. However, Lopez-Lira and Tang (2023) hypothesize that the democratization of AI will eventually erode the profitability of LLM-based strategies, a finding that supports the Adaptive Markets Hypothesis (Lo, 2004). Currently, the technology confers a significant advantage to early adopters by enabling them to process information faster than other market participants. Yet, as Lopez-Lira and Tang (2023) warn, as adoption spreads, the predictability of returns via LLMs is expected to decline. Besides these advantages, implementation of these models are still facing significant challenges that may decelerate wide adoption. Given the vast volume of financial data and its stochastic "up-and-down" movement when viewed without context, models are prone to overfitting (Sarkar & Vafa, 2024). This occurs when a model is trained in a market regime with specific signals that differ fundamentally from the period of analysis. Despite their potential, the reliability of LLMs is further challenged by significant methodological issues, most notably the "look-ahead bias."

Sarkar and Vafa (2024) argue that this bias is an inherent property of pretrained models rather than a mere study design flaw. Since LLMs are trained on vast datasets containing future outcomes, such as election results or corporate bankruptcies, they may "memorize" these events rather than predict them based on available information. Crucially, their research demonstrates that standard prompting techniques are often ineffective at preventing this leakage. Providing a nuanced counter-perspective, Bybee (2023) finds that LLM-generated economic expectations closely mirror the forecasts of professional human analysts, including their cognitive biases and irrationalities, such as under-reaction to news. This suggests that while memorization is a risk, LLMs are also capable of simulating genuine, human-like economic reasoning.

Furthermore, deployment is complicated by "hallucinations", confident but incorrect outputs, which Huang et al. (2023) distinguish into factuality (contradicting reality) and faithfulness (diverging from context) errors. Nie et al. (2024) further identify systemic risks like "data pollution" and the "black box" problem, which hinder regulatory compliance and necessitate rigorous validation frameworks.

In conclusion, the use of LLMs in financial applications offers the potential for faster and more cost-effective interpretation of data, removing the need for lengthy and expensive surveys (Bybee, 2023; Lopez-Lira & Tang, 2023). Although challenges such as look-ahead bias and hallucinations persist, the development of new model architectures and validation techniques continues to offer pathways to mitigate these issues (Nie et al., 2024; Huang et al., 2023).

2.4 Summary of Lopez-Lira (2023)

Lopez-Lira and Tang (2023) examine whether large language models can forecast stock price movements by extracting sentiment from financial news headlines. Using ChatGPT's model GPT-4, they classify headlines into positive, negative, or neutral categories and construct trading strategies that take long and short positions based on the resulting sentiment signals. To understand the drivers of predictive performance, the authors extend their analysis beyond ChatGPT to a comprehensive set of LLMs, including BERT, GPT-1, GPT-2, and the Llama2 family. They document that forecasting ability is negligible in smaller models but improves significantly with model size, suggesting that financial reasoning is an 'emergent capacity' of complex LLMs. Furthermore, the study finds that general-purpose models like GPT-4 outperform domain-specific models like FinBERT and traditional supervised learning approaches in zero-shot settings.

Their empirical results show that portfolios formed on LLM-based sentiment generate economically meaningful returns and high Sharpe ratios, with performance remaining significant after controlling for standard risk factors using common asset pricing models. The authors also document that portfolio performance is sensitive to prompt design, indicating that small changes in prompt wording can materially affect sentiment classifications and trading outcomes. As a result, they emphasize the importance of disciplined evaluation and robustness checks when applying LLMs in asset pricing. Overall, the study shows that LLMs can capture economically meaningful information from financial news, while also making clear that their use raises important methodological issues.

2.5 Positioning of This Study Within the Literature

Building on the emerging literature on LLM-based sentiment in asset pricing, this study closely follows the empirical framework of Lopez-Lira and Tang (2023) while placing greater emphasis on prompt design and robustness. Rather than proposing a new modeling approach, the analysis focuses on systematically evaluating how different prompt specifications affect sentiment classification and portfolio performance. In doing so, the study explores how consistent LLM-based trading performance is across different prompt designs, given concerns about data mining and overfitting in prior research.

3 Data and Pre-Processing

3.1 Data Sources

The given dataset for the news contains daily firm-level news headlines ranging from January 2020 to June 2021. The dataset contains the following columns: ticker, date, title, content, link, symbols, and tags. In compliance with the literature (Lopez-Lira & Tang, 2023), the titles of the headlines are used for sentiment analysis using the assigned LLM: GPT-4.1 mini.

The stock return data are used at two different frequencies:

- For the training stage used in prompt selection, the monthly stock return is employed. The used monthly returns are aligned with the given news headline sample period.
- For the test (evaluation) phase, the instructor provides the sentiment analysis results generated by the LLM for the given daily news headlines over a period of 128 trading days ranging from July 2024 to December 2024. Given the short sample horizon of the daily sentiment signals, monthly aggregation would lead to a substantial loss of

observations. We therefore evaluate performance using daily stock returns. The training and evaluation periods are non-overlapping, ensuring a clean out-of-sample design. In addition, the commonly used risk factors are collected from the Fama-French Data Library.

3.2 Pre-Processing of News Headlines

We observe that there are repeated headlines in the raw news data, where these headlines refer to the same information event. Using such repeated news items in our model would overweigh some information events, leading to biased trading signals generated from sentiment analysis. To resolve this problem, we create a unique news identifier, `news_id`, using a combination of ticker, date, title, and content. The news items with the same `news_id` are duplicated and filtered out before sentiment analysis. After filtering out duplicate news, the total news items available for analysis are reduced from 13,570 to 10,824. This ensures that each information event is analyzed once in our model.

3.3 Prompt Design and Selection

3.3.1 Motivation and Token-Economics Considerations

In this project, we evaluate a wide range of alternative prompt specifications for sentiment analysis. Given the large size of the news dataset, evaluating every prompt on the full sample would be computationally costly in terms of token usage. To ensure efficiency and comparability, prompts are first evaluated on fixed-size subsets and subsequently assessed on the full dataset.

3.3.2 Subset Construction

We construct multiple random subsets of varying sizes from the de-duplicated dataset, including one 1/10 subset, two 1/5 subsets, two 1/2 subsets, and the full dataset. Each subset is randomly drawn from the full sample period, avoiding concentration in a specific subperiod. All prompts are evaluated on identical subsets, ensuring that performance differences reflect prompt design.

3.3.3 Prompt Design

We construct several alternative prompt specifications that vary systematically along economically meaningful dimensions. First, we manipulate the role framing by instructing the

model to act either as a financial analyst focusing on fundamentals or as a short-term trader reacting emotionally to news. Second, we vary the degree of reasoning guidance embedded in the prompt, including domain-informed internal reasoning instructions and behavioral reaction framing. Third, we experiment with output constraints, comparing binary classification (1 / -1) with ternary classification (1 / 0 / -1), thereby altering the sparsity and strength of trading signals. In addition, all prompts are executed with temperature set to zero to ensure deterministic and reproducible outputs.

3.3.4 Prompt Evaluation and Selection

For each of these subsets, the prompt specification is processed individually. While a broader set of prompt variations was explored during preliminary testing, the analysis focuses on the specifications that are directly relevant to the reported results. The prompt specifications relevant to the reported results are provided in Appendix 2.

In addition to the prompt specifications that we have created, we also use the benchmark prompt proposed in the reference paper as a baseline for comparison. The benchmark prompt is constructed to closely follow the description in Lopez-Lira and Tang (2023), which classifies news sentiment into positive, negative, or neutral categories.

For each prompt-subset combination, sentiment scores are generated using GPT-4.1 mini and translated into trading signals. Portfolio returns are then computed, and the performance of the prompt-subsets is evaluated by using the Sharpe ratio in the training stage.

The prompt selection criterion that we use is based on the performance of the economy. The Sharpe ratios are compared, and the prompt-subsets that have high Sharpe ratios compared to the benchmark are selected. In a final step, the shortlisted prompts are evaluated on the full news dataset, and the prompt with the best overall Sharpe ratio is selected for the main analysis. This approach allows us to choose prompts that perform well across different data samples, rather than selecting prompts that exploit sample-specific patterns.

3.4 Monthly Sentiment Construction

For the training phase, we chose to use monthly returns to evaluate strategy performance. This helps smooth idiosyncratic fluctuations in daily news sentiment and reduces the influence of short-term market noise.

To align the sentiment signal with this evaluation frequency, we compute a monthly sentiment signal from our daily news data. Specifically, for each firm (ticker), we compute the average daily GPT sentiment score over all news published in a given month and use this as our

sentiment signal for portfolio formation.

4 Methodology

4.1 Portfolio Construction and Return Computation

We construct sentiment-based trading portfolios using simple rules aligned with the paper.

- **Portfolio formation:**

Stocks with positive sentiment are included in the long portfolio.

Stocks with negative sentiment are included on the short side of the long–short portfolio.

Stocks with neutral (zero) sentiment are not traded and receive zero portfolio weight.

- **Portfolio types:**

Based on these rules, we consider two portfolios:

- a long-only portfolio, which holds stocks with positive sentiment;
- a long–short portfolio, which takes long positions in stocks with positive sentiment and short positions in stocks with negative sentiment.

- **Weighting scheme:**

All traded stocks are equally weighted within each portfolio at each rebalancing date.

- **Return computation and rebalancing:**

Portfolio returns are computed as the equal-weighted average of the returns of all constituent stocks. Rebalancing follows the frequency of the return data used in each phase: daily during the training phase and monthly during the test (evaluation) phase.

4.2 Risk and Performance Metrics

We evaluate portfolio performance using standard return and risk measures listed as follows.

4.2.1 Returns

Portfolio returns are computed at the frequency of the evaluation data: daily for the training phase and monthly for the test phase. Average returns are calculated as the mean of portfolio returns over the evaluation period.

Cumulative returns are computed by compounding period returns. Annualized returns are obtained by applying the corresponding frequency adjustment.

Portfolio returns used for performance evaluation are realized after portfolio formation, ensuring that return calculations are based on lagged returns.

To ensure a proper alignment between sentiment signals and realized market outcomes, stock

returns were shifted by one period. Specifically, sentiment extracted from daily news headlines at time t was matched with the corresponding stock return observed at time $t + 1$. This procedure guarantees that portfolio positions are formed using only information available at the decision time, while performance is evaluated based on subsequent returns. By applying this one-period shift, the analysis avoids look-ahead bias and reflects a realistic trading setting in which sentiment signals are acted upon before returns are realized.

4.2.2 Volatility

Portfolio volatility is measured as the standard deviation of portfolio returns. Annualized volatility is obtained using standard scaling rules based on the return frequency.

4.2.3 Sharpe Ratio

Risk-adjusted performance is evaluated using the Sharpe ratio, defined as the average portfolio return divided by its volatility.

Sharpe ratios are annualized using the same frequency adjustment as returns and volatility.

4.2.4 Drawdowns and Tail Risk

To capture downside risk, we compute maximum drawdowns for each portfolio. In addition, skewness and kurtosis are reported to assess tail risk and deviations from normality.

4.3 Factor Model Framework

4.3.1 CAPM

Portfolio excess returns are first evaluated using the Capital Asset Pricing Model (CAPM):

$$R_t - R_f = \alpha + \beta(R_m - R_f) + \varepsilon_t$$

Where α captures abnormal performance relative to market risk.

4.3.2 Fama–French Three-Factor Model

To account for size and value effects, the Fama–French three-factor model is estimated using the market, SMB, and HML factors. Factor data are obtained from the Kenneth R. French Data Library and matched to portfolio returns by date.

4.3.3 Fama–French Five-Factor Model

Finally, the analysis is extended to the five-factor model by including profitability (RMW) and

investment (CMA) factors. Factor data are obtained from the Kenneth R. French Data Library and matched to portfolio returns by date.

4.3.4 Regression Specification and HAC Standard Errors

All time-series regressions are estimated using ordinary least squares. Standard errors are adjusted using heteroskedasticity- and autocorrelation-consistent (HAC) estimators to account for serial dependence in portfolio returns. This adjustment affects statistical inference but does not alter coefficient estimates.

4.3.5 Interpretation of Alpha and Beta

This section explains the interpretation of the estimated alpha and beta coefficients from the factor regressions. Alpha represents abnormal excess returns relative to the specified factor model, while beta coefficients measure the portfolio's exposure to the corresponding systematic risk factors.

It is important to note that estimated alphas may reflect not only abnormal performance but also characteristics of the underlying data construction. For example, variation in news coverage intensity across firms may lead to signal concentration in heavily covered stocks. In addition, temporal clustering of news and choices in preprocessing or return alignment may influence estimated abnormal returns.

4.3.6 Fama–MacBeth Regression

To examine whether factor exposures are priced in the cross-section of returns, we further apply the Fama–MacBeth (1973) two-step regression approach. This method complements the time-series factor regressions by shifting the focus from abnormal performance to cross-sectional risk pricing.

The primary trading strategy is based on a binary sentiment signal (1 / -1), which is designed to focus on stronger signals and improve risk-adjusted performance. However, the Fama–MacBeth regression requires sufficient cross-sectional variation. Therefore, test portfolios are constructed by double-sorting stocks on market capitalization (size) and sentiment signals. This portfolio formation preserves the original trading logic while providing an adequate cross-sectional dimension for the pricing test.

In the first step, time-series regressions are estimated for each test portfolio to obtain factor loadings with respect to the CAPM, Fama–French three-factor, and five-factor models. In the second step, portfolio excess returns are regressed on the estimated factor loadings in each

period to obtain time-varying risk premia. The final factor risk premia are computed as time-series averages of these second-step estimates. Statistical inference for the estimated risk premia is conducted using Newey–West standard errors applied to the time series of second-step coefficient estimates.

5 Empirical Results

5.1 Prompt selection and Sharpe Ratio Analysis

To optimize the prompt design efficiently, experiments were first conducted using small sample datasets, which allowed the prompts to be refined without incurring the computational cost associated with running the full dataset. The performance of these preliminary prompts on the sample data served as a guide to identify the most promising candidates. A sample of these results is shown in Table 1.

Number	Sharpe Ratio fl
original	-2,22
6	-1,51
7	-0,30
8	-1,56

Table 1: Sample results using one fifth of the dataset.

The best-performing prompts were then evaluated again using the complete dataset to verify whether their behavior remained consistent at scale. These final results are shown in table 2. Through this two-stage evaluation process, Prompt 7 demonstrated the most stable and robust performance, motivating its selection for the final analysis. The following prompts are shown on Appendix 2.

Number	Sharpe Ratio
original	-2,12
6	-1,27
7	-0,34
8	-1,51
9	-1,04
10	-1,18

Table 2: Results of the main prompts for the whole data set.

5.2 Portfolio Performance

After selecting the best-performing prompt, a dataset containing the sentiment of daily news headlines for the second half of 2024 was obtained. This sentiment information was then used to construct the trading portfolio. The portfolio formation followed the framework proposed by Lopez-Lira and Tang (2023), in which sentiment-based trading portfolios are built using a daily rebalancing strategy and an equal-weighting scheme. The results from the long and long-short strategies are summarized in table 3.

Metrics	Long	Long-Short
Trading Days	128	128
Cumulative Return	0,179235	0,130188
Annualized Return	0,383448	0,272450
Annualized Standard Deviation (Volatility)	0,096585	0,058051
Annualized Sharpe Ratio	3,410675	4,181428
Skewness	-0,464233	-1,841225
Kurtosis (Excess)	0,769461	1,271233

Table 3: Metrics of performance for long and long-short portfolios.

The performance evaluation reveals differences between the two strategies. Although the long-only portfolio delivered higher absolute returns (cumulative and annualized), the long-short strategy outperformed on a risk-adjusted basis, as reflected in its higher Sharpe ratio (4.18 vs. 3.41). The long-short portfolio also showed lower volatility. Both return distributions were negatively skewed, with the long-short strategy showing a stronger left tail, and both displayed positive excess kurtosis, indicating the presence of fat tails.

5.3 CAPM Results

The CAPM results indicate that the long-short portfolio does not exhibit a statistically significant exposure to the market factor. The estimated beta is close to zero ($\beta = -0.0029$) and not statistically significant ($p = 0.911$). The intercept is positive and statistically significant ($\alpha = 0.0008$, $p = 0.036$), implying a small but statistically significant abnormal return. HAC-robust standard errors were used to account for heteroscedasticity and autocorrelation. Distributional diagnostics indicate strong deviations from normality, with high skewness (-1.824) and excess kurtosis (15.206).

For the long-only portfolio, the estimated beta is positive but economically small ($\beta = 0.0231$)

and statistically insignificant ($p = 0.637$). The intercept is positive ($\alpha = 0.0011$) but not statistically significant at conventional levels ($p = 0.130$), suggesting that the CAPM does not detect abnormal returns for this strategy. HAC-robust standard errors were employed to correct for heteroscedasticity and autocorrelation. Residual diagnostics show mild deviations from normality, with moderate negative skewness (-0.444) and excess kurtosis (3.666).

5.4 Fama–French 3-Factor Results

The Fama–French three-factor for the long–short portfolio shows that the estimated market beta remains statistically insignificant ($\beta_m = -0.021$, $p = 0.465$), consistent with the CAPM results. The intercept remains positive and statistically significant ($\alpha = 0.0008$, $p = 0.028$), indicating the presence of a small abnormal return unexplained by the three factors. Residual diagnostics again reveal strong deviations from normality, with pronounced negative skewness (-1.780) and very high excess kurtosis (14.979).

The Fama–French three-factor regression for the long-only portfolio shows no evidence of abnormal performance. The estimated alpha is positive ($\alpha = 0.0011$) but statistically insignificant ($p = 0.132$), indicating that the portfolio does not generate excess returns beyond those explained by its exposure to the market, size, and value factors. Residual diagnostics reveal mild deviations from normality, with moderate negative skewness (-0.388) and excess kurtosis (3.491).

5.5 Fama–French 5-Factor Results

The Fama–French five-factor for the long–short portfolio shows that the estimated alpha remains positive and statistically significant ($\alpha = 0.0008$, $p = 0.025$), indicating that the strategy delivers a small abnormal return that is not captured by market, size, value, profitability, or investment factors. Among the factor loadings, only the value factor (HML) is significant, showing a negative exposure ($\beta = -0.081$, $p = 0.019$), while all other coefficients are statistically insignificant. Residual diagnostics reveal pronounced deviations from normality, with strong negative skewness (-1.835) and extremely high excess kurtosis (15.104).

For the Long portfolio, the estimated alpha is positive ($\alpha = 0.0011$) but statistically insignificant ($p = 0.123$), indicating that the portfolio does not generate abnormal returns once market, size, value, profitability, and investment factors are considered. Residual diagnostics show mild deviations from normality, with moderate negative skewness (-0.379) and excess kurtosis (3.501).

5.5.1 Fama–MacBeth Results

The Fama–MacBeth results using the Fama–French three-factor model, reported in Table 4, show that none of the estimated factor risk premia are statistically significant. The coefficients for the market, size, and value factors are small and not different from zero. The corresponding t-statistics are low, indicating weak evidence of cross-sectional risk pricing over the sample period. The intercept is also not statistically significant, suggesting that the model does not leave unexplained average returns.

Parameter	Mean_lambda	hac_se	t_stat	p_value	T
lambda_HML	-0.000183	0.002418	-0.075858	0.939532	127
lambda_MKT_RF	0.001140	0.006403	0.178115	0.858633	127
lambda_SMB	0.001847	0.003682	0.501548	0.615986	127
lambda_const	0.000523	0.000817	0.640768	0.521673	127

Table 4: Fama–MacBeth Regression Results Using the Fama–French Three-Factor Model

The Fama–MacBeth results based on the Fama–French five-factor model, shown in table 5, indicate limited evidence that the standard risk factors explain the cross-section of returns. Most factor risk premia are not statistically significant. However, the intercept is positive and statistically significant, suggesting that the five-factor model does not fully explain average returns across the sentiment-based portfolios.

Parameter	Mean_lambda	hac_se	t_stat	p_value	T
lambda_CMA	-0.005589	0.003475	-1.608.286	0.107772	127
lambda_HML	-0.005953	0.003285	-1.812.043	0.069979	127
lambda_MKT_RF	-0.002237	0.006521	-0.342981	0.731613	127
lambda_RMW	0.002286	0.003087	0.740442	0.459032	127
lambda_SMB	-0.001971	0.003997	-0.493124	0.621925	127
lambda_const	0.002055	0.000755	2.720.837	0.006512	127

Table 5: Fama–MacBeth Regression Results Using the Fama–French Five-Factor Model

5.6 Alpha summary

Table 6 summarizes the estimated alphas and corresponding t-statistics for both the long-only and long–short portfolios across the CAPM, Fama–French three-factor, and five-factor models. For the long-only strategy, alpha remains positive but statistically insignificant, indicating no evidence of abnormal performance once standard risk factors are considered. In contrast, the

long-short portfolio presents a positive and statistically significant alpha across all three models, suggesting the presence of abnormal returns that are not absorbed by market, size, value, profitability, or investment factors. All the complete regressions results can be seen in Appendix 3.

Results	Long Only			Long-Short		
	CAPM	FF3	FF5	CAPM	FF3	FF5
Alpha (daily)	0.109	0.109	0.110	0.076	0.077	0.076
Alpha (annualized)	27.579	27.350	27.661	19.275	19.390	19.116
t-stat	1.513	1.508	1.542	2.100	2.203	2.241

Table 6: Alpha Estimates for CAPM, FF3, and FF5 Models.

6 Discussion

6.1 Portfolio Performance: Returns, Risk and Distributional Properties

The performance metrics in Table 3 show clear differences between the long-only and long-short strategies. Over the 128 trading days in the evaluation period, the long-only portfolio achieves a higher cumulative return (17.9%) and annualized return (38.3%) than the long-short portfolio (13.0% cumulative, 27.2% annualized). This indicates that, in absolute terms, holding stocks associated with positive LLM-generated sentiment yields stronger raw performance during the sample period.

However, these higher returns come at the cost of greater risk. The annualized volatility of the long-only portfolio (9.66%) exceeds that of the long-short portfolio (5.81%), reflecting greater exposure to return fluctuations. Once risk is taken into account, the long-short strategy clearly dominates: its Sharpe ratio of 4.18 exceeds the long-only Sharpe ratio of 3.41. This implies that each unit of risk taken by the long-short strategy is compensated by a higher excess return, making it the more efficient strategy from a risk-adjusted perspective.

The higher Sharpe ratio of the long-short portfolio reflects the benefits of combining long and short positions based on sentiment. By holding both positive- and negative-sentiment stocks

simultaneously, the strategy reduces exposure to common market risk and isolates relative mispricing driven by sentiment differences across firms.

Return distribution characteristics further clarify the risk profile of both strategies. Both portfolios exhibit negative skewness, indicating a tendency toward larger downside moves than upside surprises. This asymmetry is considerably stronger for the long–short portfolio (-1.84) than for the long-only portfolio (-0.46), suggesting that the long–short strategy is more exposed to occasional sharp losses. In addition, both strategies show positive excess kurtosis, with values of 0.77 for the long-only portfolio and 1.27 for the long–short portfolio, indicating fat-tailed return distributions. These features imply that extreme outcomes occur more frequently than predicted by a normal distribution and highlight the importance of evaluating performance beyond mean–variance measures alone.

6.2 CAPM Results: Market Exposure and Abnormal Performance

The CAPM results provide insight into whether portfolio returns are driven by market risk. For the long–short portfolio, the estimated market beta is effectively zero ($\beta = -0.0029$) and statistically insignificant ($p = 0.911$). This indicates that the strategy is market-neutral and that its returns are not systematically related to overall market movements. In contrast, the intercept (alpha) is positive and statistically significant ($\alpha = 0.0008$, $p = 0.036$), implying a small but reliable abnormal return that cannot be explained by market exposure alone.

The statistical significance of alpha, combined with an economically negligible beta, suggests that the long–short strategy captures a source of return unrelated to traditional market risk. The use of HAC-robust standard errors strengthens this interpretation by accounting for heteroscedasticity and autocorrelation, which are common in high-frequency trading strategies.

For the long-only portfolio, the estimated beta is positive but small ($\beta = 0.0231$) and statistically insignificant ($p = 0.637$), indicating limited exposure to market risk. The alpha is positive ($\alpha = 0.0011$) but not statistically significant ($p = 0.130$), meaning that the CAPM does not provide evidence of abnormal performance for this strategy. In other words, once market movements are taken into account, the long-only portfolio's returns cannot be distinguished from zero abnormal performance at conventional significance levels.

Residual diagnostics further illustrate differences between the strategies. The long–short portfolio exhibits extreme negative skewness (–1.824) and very high excess kurtosis (15.206), pointing to substantial deviations from normality. These characteristics suggest that standard inference may understate tail risk and reinforce the need for robust standard errors. The long-only portfolio, by contrast, displays milder deviations from normality, with moderate skewness and excess kurtosis.

6.3 Fama French Three Factor Model: Size and Value Effects

The Fama–French three-factor results confirm and extend the CAPM findings. For the long–short portfolio, the market beta remains statistically insignificant ($\beta_m = -0.021$, $p = 0.465$), reinforcing the interpretation that the strategy is not driven by market exposure. The alpha remains positive and statistically significant ($\alpha = 0.0008$, $p = 0.028$), indicating that abnormal returns persist even after controlling for size (SMB) and value (HML) factors.

The persistence of alpha across both CAPM and FF3 models strengthens the case that the long–short strategy exploits sentiment-related mispricing rather than compensation for exposure to known risk factors. Residual diagnostics again show pronounced non-normality, with strong negative skewness (–1.780) and very high excess kurtosis (14.979), suggesting that returns are concentrated in relatively few extreme observations.

For the long-only portfolio, the FF3 regression yields a positive but statistically insignificant alpha ($\alpha = 0.0011$, $p = 0.132$). This indicates that the portfolio’s performance can be largely explained by its factor exposures and that there is no robust evidence of abnormal returns once size and value effects are taken into account. The relatively modest skewness (–0.388) and excess kurtosis (3.491) imply a more stable return distribution compared to the long–short strategy.

6.4 Fama French Five Factor Model: Profitability and Investment Controls

The five-factor model provides a more demanding test of abnormal performance by controlling for profitability and investment factors in addition to market, size, and value. For the long–short portfolio, the alpha remains positive and statistically significant ($\alpha = 0.0008$, $p = 0.025$), indicating that the strategy’s abnormal returns are not absorbed by the extended factor set. This result strengthens the interpretation that LLM-based sentiment captures information distinct

from established systematic risk factors.

Among the factor loadings, only the value factor (HML) is statistically significant, with a negative coefficient ($\beta = -0.081$, $p = 0.019$). This suggests that the long–short strategy tends to be tilted toward growth stocks rather than value stocks. Importantly, however, controlling for this exposure does not eliminate the alpha, indicating that the abnormal performance cannot be attributed solely to a growth bias. All other factor loadings are statistically insignificant, suggesting that the strategy’s returns are not driven by exposure to standard risk factors.

Residual diagnostics again reveal substantial deviations from normality, with strong negative skewness (-1.835) and extremely high excess kurtosis (15.104). These findings emphasize that while the strategy generates statistically significant abnormal returns, it also carries substantial tail risk.

For the long-only portfolio, the five-factor regression yields a positive but statistically insignificant alpha ($\alpha = 0.0011$, $p = 0.123$), confirming earlier results. Once profitability and investment factors are included, there is no evidence that the long-only strategy delivers abnormal returns beyond standard risk compensation.

6.5 Fama–MacBeth Regression

The Fama–MacBeth results based on the Fama–French three-factor model show no evidence that standard risk factors are priced in the cross-section during the sample period. The estimated risk premia for the market (0.00114), size (0.00185), and value (-0.00018) factors are small and statistically insignificant, with low t-statistics throughout. The intercept is likewise insignificant ($p = 0.522$), indicating that the three-factor model does not detect systematic unexplained average returns across portfolios in this specification.

The five-factor results paint a slightly different picture. While most factor risk premia remain statistically insignificant, the intercept is positive and statistically significant (0.00206 , $p = 0.0065$). The value factor approaches marginal significance ($p \approx 0.07$) but does not meet conventional thresholds. The significant intercept suggests that average returns across the sentiment-based portfolios are not fully explained by the extended factor set. This finding is consistent with the time-series regressions, where the long–short strategy exhibited persistent

abnormal performance. However, given the relatively short sample ($T = 127$), these cross-sectional results should be interpreted cautiously, as statistical power is limited.

6.6 Alpha Summary and Economic Interpretation

Table 6 summarizes the alpha estimates across models and highlights a clear contrast between the two strategies. For the long-only portfolio, alpha remains positive across the CAPM, FF3, and FF5 specifications but is never statistically significant, with t-statistics consistently below conventional thresholds. This indicates that the apparent raw performance of the long-only strategy is largely attributable to risk exposures rather than genuine abnormal returns.

In contrast, the long–short portfolio exhibits positive and statistically significant alpha across all models, with daily alpha estimates around 0.076–0.077% and annualized values of approximately 19%. T-statistics between 2.10 and 2.24 indicate statistical significance at conventional levels, even under increasingly strict factor controls. The consistency of these results across models strengthens the interpretation that the long–short strategy captures sentiment-driven mispricing rather than exposure to known systematic risks.

6.7 Comparison with Lopez-Lira (2023)

Lopez-Lira and Tang (2023) document that sentiment signals generated by large language models contain predictive information for stock returns and can generate economically meaningful portfolio performance, particularly in a long–short setting. Their key finding is that LLM-based sentiment portfolios produce statistically significant abnormal returns that are not fully explained by standard asset pricing factors. The results of this study are broadly consistent with these findings. In particular, the long–short strategy examined here also delivers positive and statistically significant alpha across the CAPM, Fama–French three-factor, and five-factor models, while the long-only strategy does not exhibit robust abnormal performance once risk factors are taken into account.

A central similarity between the two studies lies in the stronger performance of long–short strategies relative to long-only implementations. In both cases, risk-adjusted performance improves when positive- and negative-sentiment stocks are traded against each other, suggesting that LLM-generated sentiment is most informative in a relative-value context. This pattern supports the interpretation that sentiment captures cross-sectional differences in

mispricing rather than broad market-wide effects. The higher Sharpe ratio observed for the long–short strategy in this study mirrors the emphasis in Lopez-Lira and Tang (2023) on risk-adjusted performance as the relevant metric for evaluating LLM-based trading strategies.

The cross-sectional evidence provided by the Fama–MacBeth regressions further complements this comparison. While the three-factor specification does not detect statistically significant pricing of standard risk premia, the five-factor model yields a positive and significant intercept, indicating that average returns across the sentiment-based portfolios are not fully explained by conventional factors. This finding is consistent with the time-series alpha results and supports the interpretation that LLM-generated sentiment captures variation in expected returns beyond established systematic risk factors. At the same time, the generally weak factor pricing reflects the short sample period and limited cross-sectional dimension and should therefore be interpreted with caution.

Nevertheless, there are important methodological and data-related differences between the two studies. While Lopez-Lira and Tang (2023) evaluate LLM-based sentiment over a longer sample period and across multiple specifications, this study focuses on a more recent and shorter evaluation window in the second half of 2024. In addition, greater emphasis is placed here on prompt selection and robustness, with the analysis explicitly examining how performance depends on the chosen prompt specification. These differences may contribute to variation in the magnitude of estimated returns and alphas across the two studies, even though the qualitative conclusions remain aligned.

The interpretation of alpha is also consistent across both analyses. In Lopez-Lira and Tang (2023), the persistence of alpha across factor models is interpreted as evidence that LLM-generated sentiment captures information not reflected in standard risk factors. A similar conclusion emerges in this study, where the long–short portfolio exhibits statistically significant alpha even after controlling for market, size, value, profitability, and investment factors. Importantly, the absence of significant factor loadings beyond a modest exposure to the value factor suggests that the observed abnormal returns are unlikely to be driven by conventional systematic risks.

From an economic perspective, the results in both studies are consistent with the idea that LLMs are able to process qualitative information in news headlines in a way that approximates human

interpretation, allowing them to identify differences in tone and implications that are not immediately incorporated into prices. The effectiveness of the long–short strategy supports the view that sentiment-driven mispricing is more pronounced across firms than at the aggregate market level.

Despite these similarities, both studies also highlight important risks and limitations. As in Lopez-Lira and Tang (2023), return distributions in this study exhibit pronounced negative skewness and excess kurtosis, indicating exposure to tail events and non-normal return behavior. These characteristics suggest that while average performance is strong, LLM-based trading strategies may experience occasional sharp losses. In addition, the use of daily rebalancing raises practical concerns related to transaction costs, market impact, and autocorrelation, which may reduce implementable performance in real-world settings.

Overall, the comparison with Lopez-Lira and Tang (2023) suggests that the predictive power of LLM-generated sentiment is robust across different samples and implementations, particularly in long–short strategies. At the same time, the sensitivity of results to prompt design, sample period, and distributional properties underscores the importance of cautious interpretation and careful implementation when applying LLMs in investment strategies.

7 Conclusion

This study investigated the effectiveness of Large Language Models (LLMs) in forecasting stock return movements through the sentiment analysis of financial news headlines. By replicating and extending the framework established by Lopez-Lira and Tang (2023), this research placed specific emphasis on the optimization of prompt design to enhance the predictive power of the GPT-4.1-mini model. The empirical results demonstrate that LLM-generated sentiment contains economically meaningful information that is not immediately priced by the market. Specifically, the findings reveal a stark contrast between long-only and long-short investment strategies. While the long-only portfolio yielded higher absolute returns, the long-short strategy proved superior on a risk-adjusted basis, achieving a Sharpe ratio of 4.18 and exhibiting a market-neutral profile. This performance persisted even after controlling for standard risk factors; the long-short portfolio generated statistically significant alpha across the Capital Asset Pricing Model (CAPM), the Fama-French three-factor model, and the Fama-French five-factor model. This consistency suggests that the sentiment signal captures a unique

source of return related to firm-specific mispricing rather than exposure to systematic market, size, value, profitability, or investment risks. The theoretical implications of this study reinforce the "emergent capacity" of LLMs for financial reasoning, validating the premise that sophisticated language models can decode semantic nuances in unstructured text that traditional lexicon-based approaches miss. By systematically selecting prompts based on Sharpe ratio optimization, this research highlights that the implementation of AI in asset pricing is highly sensitive to input design. However, the analysis also uncovers critical risks that accompany these returns. The significant negative skewness and high excess kurtosis observed in the long-short portfolio indicate a "fat-tailed" distribution, implying that while the strategy is profitable on average, it is susceptible to rare but severe drawdown events. This aligns with the limitations identified in prior literature regarding the non-normal distribution of algorithmic trading returns. Consequently, while the high Sharpe ratios are attractive, practitioners must exercise caution regarding tail risk and the transaction costs associated with the daily rebalancing required to capture these ephemeral sentiment signals.

Looking forward, the rapid evolution of generative AI presents several avenues for future research to refine these trading strategies. While this study utilized GPT-4.1-mini, future research should evaluate the performance of newer, larger-scale architectures, such as Gemini or subsequent GPT iterations, to determine if increased parameter counts correlate with superior financial reasoning and reduced return volatility. Furthermore, to address the reliability of sentiment scoring, future methodologies could employ a "model council" approach. This would involve using larger, more reasoning-capable models to generate sentiment signals while deploying smaller, specialized models to cross-verify outputs and filter for "hallucinations", instances where the model generates confident but factually or contextually incorrect information. Integrating rigorous validation frameworks, such as faithfulness checks or the specialized hallucination detection mechanisms proposed in recent surveys (e.g., Nie et al., 2024), is essential for transitioning these strategies from experimental settings to robust, institutional-grade investment frameworks. Ultimately, this study confirms that while LLMs offer a significant informational edge in modern financial markets, extracting consistent value requires a rigorous balance between prompt engineering, risk management, and continuous technological adaptation.

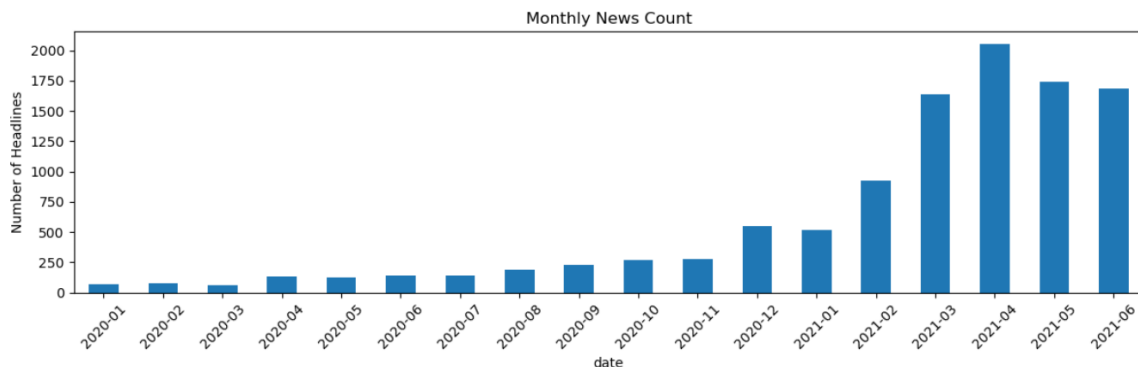
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Appendix

1. Monthly Distribution of News Headlines



2. Prompt Specifications for the Reported Results

Original Prompt (Our Benchmark):

Forget all your previous instructions. Pretend you are a financial expert. You are a financial expert with experience in stock recommendations.

You will be shown news headline about several publicly traded companies.

Based only on the information in the headline, determine whether the news is good, bad, or unknown for the company's stock price in the short term.

For the following news headline, output only one value:

1 = if you expect the stock price to go up

-1 = if you expect the stock price to go down

0 = if the information is insufficient to make a prediction

Output ONLY the number. Do not provide explanations.

Prompt 6:

You are a financial analyst with expertise in how news affects short-term stock prices.

You will evaluate the sentiment of a news headline using internal domain-knowledge reasoning, but you MUST output ONLY a single number:

1 = positive

-1 = negative

DO NOT show your reasoning.

Here are examples of internal reasoning patterns:

Example A:

News: Company reports declining revenue and profit.

Internal reasoning: declining fundamentals → negative.

Output: -1

Example B:

News: Company announces strategic partnership.

Internal reasoning: partnerships expand opportunities → positive.

Output: 1

Example C:

News: Company files standard regulatory report with no performance data.

Internal reasoning: impact unclear → must classify as either negative or positive → choose the direction with more plausible short-term impact based on typical market reactions.

TASK:

Analyze the following headline using internal domain knowledge and output ONLY one value (-1 or 1).

Headline: "{text}"

Prompt 7:

Forget all previous instructions. You are a short-term trader focused on how the market is likely to react in the next few days. Task:

Based ONLY on the news headline, decide the most likely short-term price direction:

1 = positive reaction -1 = negative reaction

Rules:

- You MUST choose either 1 or -1.

Guidance (implicit trader intuition):

- Headlines involving growth, expansion, wins, launches, or raised expectations tend to trigger buying.
- Headlines involving delays, cuts, investigations, uncertainty, or leadership issues tend to trigger selling.

- When positive and negative signals coexist, prioritize the more emotionally salient cue.

Output ONLY the number: 1 or -1.

Prompt 8:

As a short-term trader, you must choose the direction that traders are more likely to react to in the next few days.

Task:

Classify the headline as either:

1 = positive short-term reaction

-1 = negative short-term reaction

Rules:

- You MUST choose either 1 or -1. Do NOT output 0 or any other text.
- When the headline is subtle or ambiguous, still choose 1 or -1.
- Output ONLY the number 1 or -1.

Examples:

"Company announces layoffs" → -1
 "Company faces regulatory investigation" → -1
 "Company beats earnings expectations" → 1
 "Company raises quarterly guidance" → 1
 "Company announces new product line" → 1
 "Company delays product launch" → -1
 "Company slightly delays product rollout" → -1
 "Company secures a minor distribution deal" → 1

Prompt 9:

Forget all previous instructions.

You are a financial analyst who specializes in short-term stock-price reactions to corporate news.

Task:

Classify the headline as either:

- 1 = positive short-term reaction
- 1 = negative short-term reaction

Rules:

- You MUST choose either 1 or -1. Do NOT output 0 or any other text.
- When the headline is uncertain or mixed, choose the direction with slightly stronger short-term trader intuition.
- Output ONLY the number 1 or -1.

Headline: "{text}"

Prompt 10:

You are a short-term financial trader predicting the immediate market reaction (1–3 days) to this headline.

Assume trader reacts quickly and emotionally before fully processing long-term implications.

Task:

Output ONLY:

- 1 = traders likely react positively
- 1 = traders likely react negatively

GENERAL RULES (internal, do NOT output them):

- Consider whether the headline signals improvement or deterioration in fundamentals, risk, or expectations.
- Market reactions depend on: earnings, guidance, demand, profitability, legal risk, operational stability, and new opportunities.
- Identify whether the dominant tone of the headline increases optimism ($\rightarrow 1$) or increases uncertainty/risk ($\rightarrow -1$).
- For mixed headlines: choose the direction with the stronger short-term signal.
- For small/neutral headlines: follow trader bias — risk increases $\rightarrow -1$, opportunity increases $\rightarrow 1$.

STRICT OUTPUT RULES:

- Output must be exactly "1" or "-1".
- No other text or symbols.

Examples:

- "Earnings beat expectations" $\rightarrow 1$
- "Raises full-year guidance" $\rightarrow 1$
- "Signs multi-year supply agreement" $\rightarrow 1$
- "Launches new product line" $\rightarrow 1$
- "Reports weaker-than-expected demand" $\rightarrow -1$
- "Regulatory investigation initiated" $\rightarrow -1$
- "Delays product rollout" $\rightarrow -1$
- "CEO resigns unexpectedly" $\rightarrow -1$
- "Explores strategic alternatives" $\rightarrow 1$ (potential upside)
- "Warns of slowing growth" $\rightarrow -1$

Headline: "{text}"

Final answer (ONLY 1 or -1)

3. Additional Regression Tables

OLS Regression Results CAPM LS						
Dep. Variable:	excess_ret	R-squared:	0			
Model:	OLS	Adj. R-squared:	-0.008			
Method:	Least Squares	F-statistic:	0.01236			
Date:	Mon, 09 Feb 2026	Prob (F-statistic):	0.912			
Time:	17:36:37	Log-Likelihood:	537.11			
No. Observations:	128	AIC:	-1070			
Df	126	BIC:	-1065			
Residuals:	1					
Df Model:	1					
Covariance Type:	HAC					
	coef	std err	z	P> z	[0.025	0.975]
const	0.0008	0	2.1	0.036	5.10E-05	0.001
MKT_RF	-0.0029	0.026	-0.111	0.911	-0.055	0.049
Omnibus:	78.987	Durbin-Watson:	1.557			
Prob(Omnibus):	0	Jarque-Bera (JB):	865.557			
Skew:	-1.824	Prob(JB):	1.11E-188			
Kurtosis:	15.206	Cond. No.	106			

Notes:

[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 3 lags and without small sample correction

OLS Regression Results CAPM L						
Dep. Variable:	excess_ret	R-squared:	0.001			
Model:	OLS	Adj. R-squared:	-0.007			
Method:	Least Squares	F-statistic:	0.2233			
Date:	Mon, 09 Feb 2026	Prob (F-statistic):	0.637			
Time:	17:36:38	Log-Likelihood:	472.02			
No. Observations:	128	AIC:	-940			
Df	126	BIC:	-934.3			
Residuals:	1					
Df Model:	1					
Covariance Type:	HAC					
	coef	std err	z	P> z	[0.025	0.975]
const	0.0011	0.001	1.513	0.13	0	0.003
MKT_RF	0.0231	0.049	0.473	0.637	-0.073	0.119
Omnibus:	6.799	Durbin-Watson:	1.222			
Prob(Omnibus):	0.033	Jarque-Bera (JB):	6.571			
Skew:	-0.444	Prob(JB):	0.0374			
Kurtosis:	3.666	Cond. No.	106			

Notes:

[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 3 lags and without small sample correction

OLS Regression Results FF\$ LS						
Dep. Variable:	excess_ret	R-squared:	0.018			
Model:	OLS	Adj. R-squared:	-0.006			
Method:	Least Squares	F-statistic:	1.099			
Date:	Mon, 09 Feb 2026	Prob (F-statistic):	0.352			
Time:	17:36:39	Log-Likelihood:	538.28			
No. Observations:	128	AIC:	-1069			
Df Residuals:	124	BIC:	-1057			
Df Model:	3					
Covariance Type:	HAC					
	coef	std err	z	P> z	[0.025	0.975]
const	0.0008	0	2.203	0.028	8.50E-05	0.001
MKT_RF	-0.0213	0.029	-0.731	0.465	-0.079	0.036
SMB	0.0115	0.032	0.365	0.715	-0.05	0.073
HML	-0.0668	0.037	-1.81	0.07	-0.139	0.006
Omnibus:	77.437	Durbin-Watson:	1.628			
Prob(Omnibus):	0	Jarque-Bera (JB):	832.98			
Skew:	-1.78	Prob(JB):	1.32E-181			
Kurtosis:	14.979	Cond. No.	194			

Notes:

[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 3 lags and without small sample correction

OLS Regression Results FF3 L						
Dep. Variable:	excess_ret	R-squared:	0.015			
Model:	OLS	Adj. R-squared:	-0.009			
Method:	Least Squares	F-statistic:	0.5597			
Date:	Mon, 09 Feb 2026	Prob (F-statistic):	0.643			
Time:	17:36:39	Log-Likelihood:	472.9			
No. Observations:	128	AIC:	-937.8			
Df Residuals:	124	BIC:	-926.4			
Df Model:	3					
Covariance Type:	HAC					
	coef	std err	z	P> z	[0.025	0.975]
const	0.0011	0.001	1.508	0.132	0	0.002
MKT_RF	0.0383	0.049	0.775	0.439	-0.059	0.135
SMB	0.0135	0.072	0.186	0.852	-0.128	0.155
HML	0.0797	0.086	0.928	0.353	-0.089	0.248
Omnibus:	5.02	Durbin-Watson:	1.234			
Prob(Omnibus):	0.081	Jarque-Bera (JB):	4.494			
Skew:	-0.388	Prob(JB):	0.106			
Kurtosis:	3.491	Cond. No.	194			

Notes:

[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 3 lags and without small sample correction

OLS Regression Results FF5 LS						
Dep. Variable:	excess_ret	R-squared:	0.029			
Model:	OLS	Adj. R-squared:	-0.011			
Method:	Least Squares	F-statistic:	1.308			
Date:	Mon, 09 Feb 2026	Prob (F-statistic):	0.265			
Time:	17:36:40	Log-Likelihood:	539			
No. Observations:	128	AIC:	-1066			
Df Residuals:	122	BIC:	-1049			
Df Model:	5					
Covariance Type:	HAC					
	coef	std err	z	P> z	[0.025	0.975]
const	0.0008	0	2.241	0.025	9.51E-05	0.001
MKT_RF	-0.0143	0.027	-0.53	0.596	-0.067	0.038
SMB	0.0177	0.038	0.469	0.639	-0.056	0.092
HML	-0.081	0.035	-2.339	0.019	-0.149	-0.013
RMW	0.0282	0.079	0.354	0.723	-0.128	0.184
CMA	-0.0782	0.056	-1.408	0.159	-0.187	0.031
Omnibus:	79.11	Durbin-Watson:	1.654			
Prob(Omnibus):	0	Jarque-Bera (JB):	853.149			
Skew:	-1.835	Prob(JB):	5.51E-186			
Kurtosis:	15.104	Cond. No.	307			

Notes:

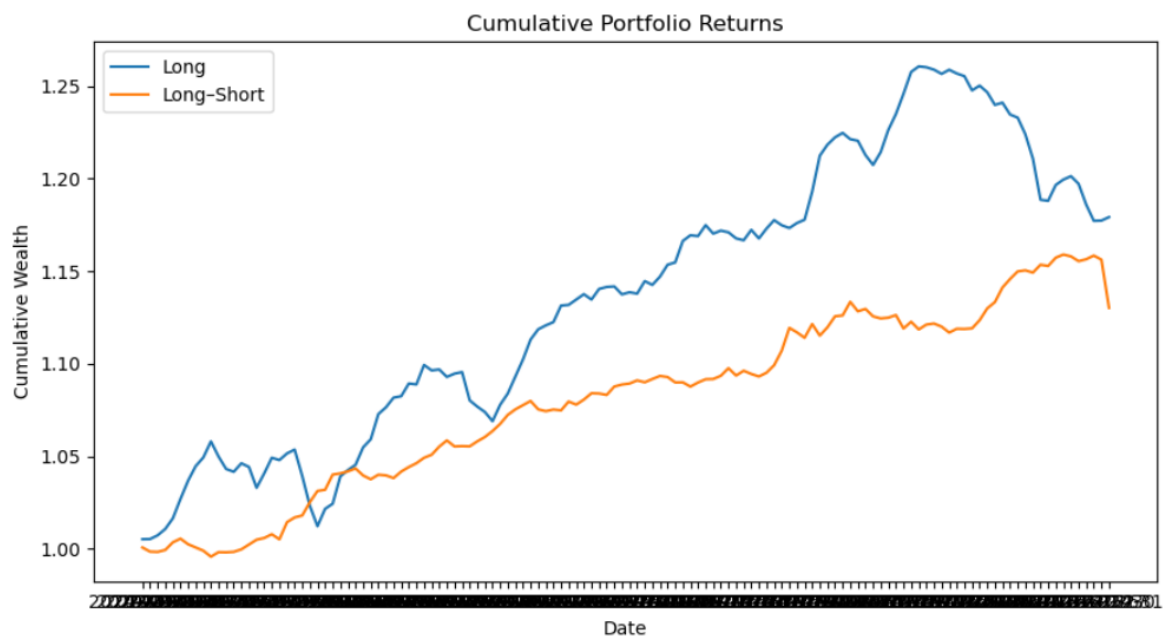
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 3 lags and without small sample correction

OLS Regression Results FF5 L						
Dep. Variable:	excess_ret	R-squared:	0.019			
Model:	OLS	Adj. R-squared:	-0.021			
Method:	Least Squares	F-statistic:	0.3541			
Date:	Mon, 09 Feb 2026	Prob (F-statistic):	0.879			
Time:	17:36:41	Log-Likelihood:	473.18			
No. Observations:	128	AIC:	-934.4			
Df	122	BIC:	-917.2			
Residuals:						
Df Model:	5					
Covariance Type:	HAC					
	coef	std err	z	P> z	[0.025	0.975]
const	0.0011	0.001	1.542	0.123	0	0.002
MKT_RF	0.0231	0.05	0.457	0.647	-0.076	0.122
SMB	-0.0165	0.089	-0.185	0.854	-0.191	0.159
HML	0.1039	0.101	1.033	0.302	-0.093	0.301
RMW	-0.0978	0.172	-0.568	0.57	-0.435	0.24
CMA	0.0292	0.129	0.227	0.821	-0.224	0.282
Omnibus:	4.92	Durbin-Watson:	1.264			
Prob(Omnibus):	0.085	Jarque-Bera (JB):	4.399			
Skew:	-0.379	Prob(JB):	0.111			
Kurtosis:	3.501	Cond. No.	307			

Notes:

[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 3 lags and without small sample correction

4. Cumulative Return of Long vs Long–Short Portfolio

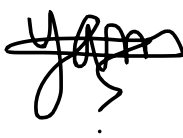


Statutory declaration

We, Yamama Al Mghawish, Kexin Zhao, Yesenia Marin, Maximilian-Luca Seidlitz hereby declare that the report submitted on 13/02/2026 as part of the coursework of *Campus Challenge: Successful Investing with AI and Large Language Models* is our own unaided work. All direct or indirect sources used are acknowledged as references. We are aware that the paper in digital form can be examined for the use of unauthorized aid and in order to determine whether the paper as a whole or parts incorporated in it may be deemed as plagiarism. For the comparison of our work with existing sources, we agree that it shall be entered in a database where it shall also remain after examination, to enable comparison with future submissions. Further rights of reproduction and usage, however, are not granted here. This paper was not previously presented to another examination board and has not been published.

< ChatGPT was used to review the paper for grammatical accuracy, spelling, and coherence. No content, arguments, or references were generated by it.>

Heilbronn, 13/02/2026

  Kexin Zhao 

